

One small-field blocking step for $SU(3)$, part 1: the Gaussian fluctuation integral with block averaging, with explicit constants

This opens the line set in the research-directions note: one Euclidean blocking step of the small-field renormalization for $SU(3)$ with every constant written down. Part 1 is the Gaussian level, where the Wilson action is expanded to second order and the fluctuation integral is exact. Four things are established. **(i)** A block averaging W of link variables that intertwines gauge transformations, $Wd = \bar{d}W_0$, so that the coarse field is a gauge field. **(ii)** In Feynman gauge the quadratic Wilson form separates into $4 \times 8 = 32$ identical scalar problems, one per direction and colour, and the fluctuation integral is the conditional law of a massless free field given its tube averages. **(iii)** A block Poincaré inequality with an explicit constant,

$$\langle f, -\Delta f \rangle \geq \frac{4}{9} \|f\|^2 \quad \text{on tube-mean-zero } f,$$

so the fluctuation covariance is at most $\frac{9}{4}g^2$ and the root-mean-square fluctuation of one link angle is at most $\frac{3}{2}g$; the alternating modes show the true constant lies in $[\frac{4}{9}, 4]$. **(iv)** The coarse quadratic form is the Wilson form at the same g^2 for slowly varying fields, the classical scale invariance of four dimensions, and its correction and the fluctuation propagator decay exponentially. For the decay rate the generic Combes–Thomas argument gives about 7×10^{-4} per lattice unit against a true rate of order one: **the weak side’s sharpness problem begins at the Gaussian propagator**, and the block random-walk expansion is the defined next piece. Units: $\hbar$ and c enter only through $S_E/\hbar$, and all lattice quantities are dimensionless with $a = 1$ where not shown. Constants explicit; nothing promoted.

1. Conventions

Fine lattice $\mathbb{Z}^4$, spacing a ; link angles $\theta_\ell \in \mathfrak{su}(3)$, $U_\ell = e^{i\theta_\ell}$, so $\theta = aA$ with A the gauge potential and g dimensionless. With $\text{tr } T^a T^b = \frac{1}{2}\delta^{ab}$,

$$\frac{S_W}{\hbar} = \frac{2}{g^2} \sum_p (N - \text{Re tr } U_p) = \frac{1}{2g^2} \sum_p \sum_{a=1}^8 ((d\theta)_p^a)^2 + O(\theta^3),$$

where $(d\theta)_p = \theta_{x,\mu} + \theta_{x+\hat{\mu},\nu} - \theta_{x+\hat{\nu},\mu} - \theta_{x,\nu}$ is the lattice exterior derivative. The quadratic part is eight copies of the lattice Maxwell form $S_2(\theta) = \frac{1}{2g^2} \|d\theta\|^2$, and the fluctuation amplitude is of order g : a field strength η per plaquette costs $\eta^2/(2g^2)$, in agreement with the large-field bound $e^{-\eta^2/(4g^2)}$ of the lower-bound note in its continuum normalization. The cubic and quartic terms, the Haar measure correction $dU = (1 + O(\theta^2))d\theta$ and the non-abelian part of the averaging are the remainder, treated in part 2.

2. The averaging and its intertwining property

Blocks B of side 2; coarse links (B, μ) from B to $B' = B + 2\hat{\mu}$. Define, for each colour component,

$$(W\theta)_{B,\mu} = \frac{1}{16} \sum_{x \in B} (\theta_{x,\mu} + \theta_{x+\hat{\mu},\mu}),$$

the average over the sixteen sites of B of the two-link path from x to $x + 2\hat{\mu}$; and for site functions the block mean $(W_0\lambda)_B = \frac{1}{16} \sum_{x \in B} \lambda_x$.

Lemma 1 (intertwining). $Wd = \bar{d}W_0$, where $\bar{d}$ is the coarse exterior derivative.

Proof. $(Wd\lambda)_{B,\mu} = \frac{1}{16} \sum_{x \in B} [(\lambda_{x+\hat{\mu}} - \lambda_x) + (\lambda_{x+2\hat{\mu}} - \lambda_{x+\hat{\mu}})] = \frac{1}{16} \sum_{x \in B} (\lambda_{x+2\hat{\mu}} - \lambda_x) = (W_0\lambda)_{B'} - (W_0\lambda)_B. \quad \square$

So a fine gauge transformation λ moves the coarse field by the coarse gauge transformation $W_0\lambda$, and coarse Wilson loops are functions of fine gauge-invariant data. This is the property that makes the gauge fixing of Section 3 harmless for the coarse theory.

The tube of (B, μ) is the set of fine links entering the average: $T_{B,\mu} = \{(y, \mu) : y \in B \cup (B + \hat{\mu})\}$, three μ -layers of eight links with weights $w = (1, 2, 1)$, the middle layer counted from both $x = y$ and $x = y - \hat{\mu}$. Each fine link lies in one or two tubes.

3. Feynman gauge: thirty-two scalar problems

Add the gauge-fixing term $\frac{1}{2g^2} \|d^*\theta\|^2$. Then $S_2 + S_{\text{gf}} = \frac{1}{2g^2} \sum_\mu \sum_a \langle \theta_\mu^a, -\Delta \theta_\mu^a \rangle$, because $d^*d + dd^*$ is the scalar Laplacian on each component of a lattice one-form. The averaging W acts on each component separately, so the Gaussian step is 32 copies of one scalar problem: a massless free field f on $\mathbb{Z}^4$ with covariance $g^2(-\Delta)^{-1}$, conditioned on its tube averages $\bar{f}_B = \frac{1}{16} \sum_{y \in T_B} w_y f_y$ for every block B , with the tubes elongated along the direction μ of the component. The four directions are equivalent by lattice symmetry.

For gauge-invariant observables the Feynman and Landau measures agree; for the coarse field they differ by a fine pure gauge $d\lambda$, which by Lemma 1 moves the coarse field by the coarse pure gauge $\bar{d}W_0\lambda$, so the law of every coarse gauge-invariant observable is the same in both. That is what the intertwining buys.

Decomposition. Let Hf be the minimizer of $\langle f, -\Delta f \rangle$ subject to $Wf = \bar{f}$, and $\xi = f - H\bar{f}$, so $W\xi = 0$. Then, exactly,

$$\langle f, -\Delta f \rangle = \langle H\bar{f}, -\Delta H\bar{f} \rangle + \langle \xi, -\Delta \xi \rangle, \quad H = GW^T(WGW^T)^{-1}, \quad G = (-\Delta)^{-1},$$

the cross term vanishing by the variational characterization. The fluctuation ξ is Gaussian on $K = \{W\xi = 0\}$ with covariance

$$C_{\text{fl}} = g^2 [G - GW^T(WGW^T)^{-1}WG] = g^2 (P(-\Delta)P)^{-1} \Big|_K,$$

the conditional covariance of the free field given its tube averages, P the orthogonal projection onto K ; and the coarse quadratic form is

$$\bar{S}_2(\bar{f}) = \frac{1}{2g^2} \langle \bar{f}, (WGW^T)^{-1} \bar{f} \rangle = \min \left\{ \frac{1}{2g^2} \langle f, -\Delta f \rangle : Wf = \bar{f} \right\}.$$

4. The block Poincaré inequality

Proposition 1. If $\sum_{y \in T_B} w_y f_y = 0$ for every block B , then

$$\|\partial f\|^2 = \langle f, -\Delta f \rangle \geq \frac{4}{9} \|f\|^2.$$

Proof. One tube T is a $2 \times 2 \times 2 \times 3$ box of 24 sites whose graph Laplacian has smallest nonzero eigenvalue $\lambda_2 = \min(\lambda_2(P_2), \lambda_2(P_3)) = \min(2, 1) = 1$, so for the plain mean m of f on T , $\sum_T (f - m)^2 \leq \sum_{\text{bonds} \subset T} (\partial f)^2$. The weighted constraint gives $m = -\frac{1}{32} \sum_T (w_y - \bar{w})(f_y - m)$ with $\bar{w} = \frac{4}{3}$, and $\sum_T (w_y - \bar{w})^2 = 8(\frac{1}{9} + \frac{4}{9} + \frac{1}{9}) = \frac{16}{3}$, so $|m| \leq \frac{4/\sqrt{3}}{32} \|f - m\|_T$ and $24m^2 \leq \frac{1}{8} \|f - m\|_T^2$. Hence $\sum_T f^2 = \|f - m\|_T^2 + 24m^2 \leq \frac{9}{8} \sum_{\text{bonds} \subset T} (\partial f)^2$. Summing over all tubes of the direction μ : every site lies in at least one tube, and every bond lies in at most two, so $\|f\|^2 \leq \frac{9}{8} \cdot 2 \|\partial f\|^2$. $\square$

Consequences. On K the precision $P(-\Delta)P/g^2 \geq \frac{4}{9g^2}$, so

$$\|C_{\text{fl}}\| \leq \frac{9}{4} g^2, \quad \langle \xi_\ell^2 \rangle^{1/2} = C_{\text{fl}}(\ell, \ell)^{1/2} \leq \frac{3}{2} g$$

for every fine link and colour. The modes alternating in sign between consecutive μ -layers, or between consecutive sites in a transverse direction, satisfy the constraint and have $\langle f, -\Delta f \rangle = 4\|f\|^2$, so the sharp constant lies in $[\frac{4}{9}, 4]$; the value $\frac{4}{9}$ is what the tube-by-tube argument gives and is enough for part 2, whose smallness condition is on $\frac{3}{2}g$.

5. The coarse form, and the decay problem

Scale invariance. For a slowly varying coarse field, $H\bar{f}$ is the slowly varying fine field with the same values, and $\bar{S}_2$ is the fine Maxwell form evaluated on it. In the coarse angle $\theta_c = 2\bar{f}$, the natural variable of a link of length $2a$, the result is the coarse Wilson quadratic form $\frac{1}{2g^2}\|\bar{d}\theta_c\|^2$ with the same g^2 , up to corrections of relative order $\bar{k}^2 a^2$: the Maxwell action is scale invariant in four dimensions, and the Gaussian step does not run the coupling. The running appears in part 2, from the dependence of the fluctuation determinant on the background through the cubic and quartic vertices.

Decay. Both C_{fl} and the correction to the coarse form decay exponentially, because the conditioning removes the long-wavelength modes; the massless pole of G cancels exactly in $G - GW^T(WGW^T)^{-1}WG$ at $\bar{k} \rightarrow 0$. An explicit rate is the first place where the weak side loses sharpness. The generic Combes–Thomas argument on $\tilde{A} = P(-\Delta)P + \frac{4}{9}(1 - P) \geq \frac{4}{9}$, conjugated by $e^{\kappa x \cdot e}$, needs the conjugation error below $\frac{2}{9}$; the Laplacian contributes $2\sinh \kappa$, and the projection, through $(WW^T)^{-1} = (48 + 8(S + S^{-1}))^{-1}$ along μ , contributes $\|P_\kappa - P\| \leq 10\kappa + O(\kappa^2)$ multiplied by $2\|-\Delta\| \simeq 33$, so

$$\kappa \leq \frac{2/9}{2 + 330} \simeq 7 \times 10^{-4} \text{ per lattice unit,} \quad |C_{\text{fl}}(x, y)| \leq \frac{9}{2} g^2 e^{-\kappa|x-y|_\infty}.$$

The true rate is of order one per block, as the exact cancellation of the pole and the alternating-mode energies indicate; the factor of a thousand is the generic argument’s price for the non-locality of P and the size of $\|-\Delta\|$. Balaban’s propagator paper (Commun. Math. Phys. 95 (1984) 17; metadata level) obtains the order-one rate by a random-walk expansion adapted to the block structure; carrying that expansion with explicit constants is the next piece, and every downstream constant, the locality of the effective interaction and the polymer activities of part 2, inherits its rate.

6. What part 2 needs from here

The remainder $S_W - S_2$ on the small-field region, the Haar correction and the non-abelian averaging are functions of $H\bar{\theta} + \xi$ with ξ Gaussian of size $\frac{3}{2}g$ per link; their expansion is a polymer gas whose activities carry powers of g and whose locality carries the decay rate of C_{fl} . The large-field region $|\theta_\ell| \gtrsim 1$ has probability at most e^{-c/g^2} per link with c of order $\frac{2}{9}$ from the covariance bound, consistent with the lower-bound note. The deliverable of part 2 is the Kotecký–Preiss threshold g_{pert}^2 for this gas.

7. Consequence for STATE

Part 1 of the explicit small-field step is in place: an intertwining averaging, the reduction to one scalar conditional-Gaussian problem in Feynman gauge, the block Poincaré constant $\frac{4}{9}$ with fluctuation size $\frac{3}{2}g$ per link, scale invariance of the coarse form, and the decay problem located with its generic rate 7×10^{-4} against a true rate of order one. Next: the block random-walk expansion for the decay rate, then part 2.